



CMPT 295

Unit - Data Representation

Lecture 3 – Representing (encoding) integral numbers in memory
- unsigned and signed

Solution to Homework and more!

Homework

Range of possible values?

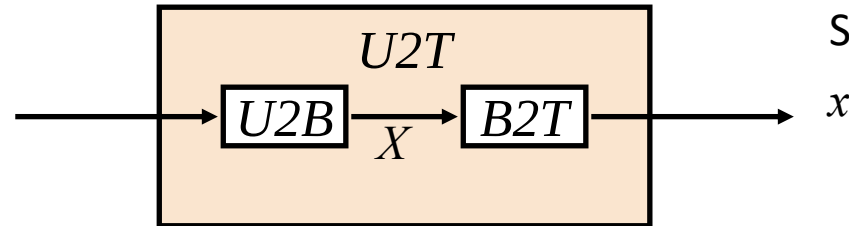
- If the context is **C** on our *target machine*, what is the range of possible values for the following **C** data types:
 - unsigned char? $[0 .. 2^8 - 1] = [0 .. 255]$
 - signed short? $[-2^{16-1} .. 2^{16-1} - 1] = [-2^{15} .. 2^{15} - 1]$
 - unsigned int? $[0 .. 2^{32} - 1]$
 - unsigned long? $[0 .. 2^{64} - 1]$
 - signed int? $[-2^{32-1} .. 2^{32-1} - 1] = [-2^{31} .. 2^{31} - 1]$
 - signed char? $[-2^{8-1} .. 2^{8-1} - 1] = [-2^7 .. 2^7 - 1] = [-128 .. 127]$
 - unsigned short? $[0 .. 2^{16} - 1]$
 - signed long? $[-2^{64-1} .. 2^{64-1} - 1] = [-2^{63} .. 2^{63} - 1]$

Converting unsigned $\leftrightarrow$ signed – Method 1 of same size (same data type)

$w = 8$

If $ux = 129_{10}$

Unsigned
 ux



Maintain Same Bit Pattern

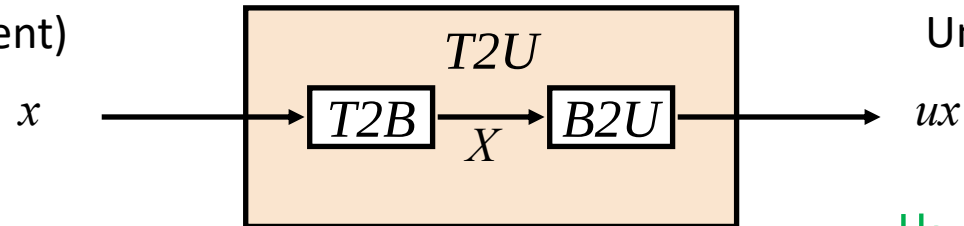
Signed (Two's Complement)
 x

then $x =$

$w = 4$

Signed (Two's Complement)
 x

If $x = -5_{10}$



Maintain Same Bit Pattern

Unsigned
 ux

then $ux = 11_{10}$

T2B(-5) using Method 1:
 $(\sim(\text{U2B}(|-5|))) + 1$
 $= (\sim(\text{U2B}(5))) + 1$
 $= (\sim(0101_2)) + 1$
 $= (1010_2) + 1$
 $= 1011_2$
 $\text{B2U}(1011_2) = 8+2+1 = 11$

- **Conclusion** - Converting between unsigned and signed numbers: Both have **same bit pattern**, however, this bit pattern may be **interpreted differently**, i.e., producing a different integral value